tvameva.ai

tvameva.ai | Proposal

The Challenge

AI-native. Outcome-anchored.

Confidential

Submitted to: Jainam Shah, UPL Limited

Submitted by: tvameva.ai

Date: April 2026

Version: 1.0

Classification: Confidential

Document Control

Version	Date	Author	Description
1.0	April 2026	tvameva.ai	Initial proposal submission

tvameva.ai Contact:

- Varada Iyengar, Chief Architect & Engagement Lead
- Website: https://tvameva.ai

UPL Limited Contact:

- Jainam Shah
- Website: https://www.upl-ltd.com/

Table of Contents

1. [Executive Summary](#1-executive-summary)
2. [Understanding of Requirements](#2-understanding-of-requirements)
3. [Proposed Solution & Architecture](#3-proposed-solution--architecture)
4. [Use Cases & Agentic Workflows](#4-use-cases--agentic-workflows)
5. [Delivery Approach](#5-delivery-approach)
6. [Team Composition](#6-team-composition)
7. [Pricing & Commercial Model](#7-pricing--commercial-model)
8. [Timeline & Milestones](#8-timeline--milestones)
9. [Risk & Mitigation](#9-risk--mitigation)
10. [Terms & Conditions](#10-terms--conditions)
11. [Cybersecurity & Data Ownership Response (Annexure-1)](#11-cybersecurity--data-ownership-response)
12. [Appendix A: Use Case Feasibility & Accuracy Matrix](#appendix-a-use-case-feasibility--accuracy-matrix)
13. [Appendix B: Platform Comparison & Selection Rationale](#appendix-b-platform-comparison--selection-rationale)

1. Executive Summary

The Challenge

UPL Limited operates 44 manufacturing plants worldwide, handling specialty chemicals and agrochemical products in one of the most regulated manufacturing environments. These plants are equipped with approximately 1,969 cameras across UPL and Superform facilities — a significant investment in physical surveillance infrastructure. Today, this infrastructure is passive. Cameras record, but they don't reason. Safety monitoring depends on human observers who cannot maintain consistent vigilance across thousands of camera feeds, around the clock, at every plant.

The result: safety incidents that AI could detect in seconds are discovered in minutes or not at all. PPE compliance is enforced through periodic audits rather than continuous monitoring. Hazardous events like fire, smoke, and chemical spillage rely on physical sensors and human senses rather than intelligent visual analysis.

What Makes This Different

tvameva.ai proposes not just a video analytics system, but a safety intelligence platform built on our InsightLens practice — our predictive and prescriptive analytics capability, extended to industrial computer vision.

Three differentiators set this proposal apart:

1. **Conversational Safety Intelligence**: Beyond predefined dashboards, UPL leadership can query safety data in natural language — "Show me PPE violations at Ankleshwar this week" — and receive prescriptive recommendations, not just reports.
2. **Agentic Delivery Model**: AI-enabled pods where agentic workflows lead execution — automated model retraining, intelligent alert tuning, continuous accuracy improvement — with human experts providing guardrails. This accelerates delivery and ensures the system improves continuously after deployment.
3. **Composable Architecture**: Edge inference (NVIDIA Metropolis) + cloud training (GCP Vertex AI) + analytics (BigQuery + Gemini). Each component is independently upgradeable. UPL is not locked into a monolithic platform.

Project Summary

Attribute	Detail
Solution	AI-powered video analytics for plant safety & security
Scope	~1,969 cameras, 16 use cases, across UPL and Superform plants
Technology	NVIDIA Metropolis (edge) + GCP Vertex AI (cloud) + Gemini (conversational analytics)
CV Partner	Wobot Intelligence (India-based, NVIDIA partner, manufacturing focus)
Hardware Partner	Dell Technologies / Advantech (NVIDIA Metropolis system builders)
Delivery Model	Agentic (AI-enabled pods) + Hybrid (remote development + on-site deployment)

Timeline	12-14 weeks to full production (pilot + rollout)
Accuracy Target	95-97% for mature use cases, 88-95% for niche use cases

Project Phasing

Phase	Scope	Duration
Phase 1: Discovery & Setup	Site surveys, camera mapping, infrastructure setup	2 weeks
Phase 2: Pilot Build & Deploy	1-2 plants, 5 priority use cases, dashboard MVP, mobile app v1	6 weeks
Phase 3: Rollout	Remaining plants, all 16 use cases, full dashboard + mobile	4-5 weeks
Phase 4: Operate & Optimize	Managed service, continuous improvement, new use cases	Ongoing
Total to Production		**12-14 weeks**

Investment Summary

Commercial Model	One-Time (INR Cr)	Annual Recurring (INR Cr)
Option A: CAPEX (One-Time + AMC)	14.5	1.3
Option B: SaaS (Per-Camera/Month)	7.5 (setup + hardware)	8.9
Option C: Perpetual License	15.5	1.6

Detailed breakdown in Section 7.

2. Understanding of Requirements

2.1 Objectives

We understand UPL's primary objective is to transform its existing camera infrastructure into an intelligent safety monitoring system. Specifically:

1. **Real-time safety monitoring** across all manufacturing plants using AI-powered video analytics
2. **95-97% accuracy** for defined use cases through continuous AI model development and fine-tuning
3. **Sub-10-second alert delivery** via SMS, WhatsApp, and email for critical safety events
4. **Enterprise dashboard** for centralized monitoring, alert lifecycle management, and cross-plant analytics
5. **Mobile application** for leadership access and on-the-go monitoring
6. **Comprehensive support** with defined SLAs for remote (4hr) and on-site (24-48hr) resolution

2.2 Scope Acknowledgment

Vendor Scope (tvameva + partners):

1. GPU hardware procurement and deployment at UPL sites (via hardware partner)
2. AI/ML model development and deployment for all 16 use cases
3. Image annotation and use case to camera mapping (ROI definition)
4. Integration with existing camera infrastructure
5. Alert notification system (SMS, WhatsApp, email)
6. Enterprise dashboard and mobile app development
7. Conversational analytics interface (value-add beyond RFP scope)
8. Ongoing support and maintenance per SLA

UPL Scope:

1. Existing camera infrastructure and access to camera feeds
2. Procurement of additional cameras recommended by tvameva
3. Site access for survey, installation, and testing
4. Network infrastructure between cameras and edge servers
5. Stakeholder availability for checkpoints and UAT

2.3 Camera Infrastructure

- **Total cameras**: ~1,969 across UPL and Superform
- **Growth accommodation**: Architecture designed to scale for 5-10% camera count changes
- **Camera assessment**: During Phase 1 (Discovery), we will assess each camera's specifications (resolution, frame rate, field of view, connectivity) and map use cases to cameras based on coverage and quality
- **Additional cameras**: Where existing cameras are insufficient for specific use cases (e.g., ANPR requires dedicated cameras at entry/exit, low-lux areas need IR/thermal cameras), we will specify requirements for UPL procurement

2.4 Use Case Mapping

All 16 use cases from the RFP are addressed. See Appendix A for the complete feasibility and accuracy matrix.

3. Proposed Solution & Architecture

3.1 Architecture Overview

The solution is built on a four-tier architecture: Edge, Cloud, Application, and Intelligence.

```
┌──────────────────────────────────────────────────────────────┐
│                    INTELLIGENCE TIER                         │
│                                                              │
│    ┌──────────────────────────────────────────────────┐      │
│    │          Conversational Analytics UI             │      │
│    │    Natural Language Querying (Gemini + BigQuery) │      │
│    │    Prescriptive Recommendations                  │      │
│    │    Cross-Plant Benchmarking & Trend Analysis     │      │
│    └──────────────────────────────────────────────────┘      │
└──────────────────────────────────────────────────────────────┘
                              │
┌──────────────────────────────────────────────────────────────┐
│                    APPLICATION TIER                          │
│                                                              │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────────┐   │
│  │ Enterprise   │  │ Mobile App   │  │ Notification     │   │
│  │ Dashboard    │  │ (React       │  │ Engine           │   │
│  │ (React)      │  │  Native)     │  │ (SMS/WA/Email)   │   │
│  └──────────────┘  └──────────────┘  └──────────────────┘   │
└──────────────────────────────────────────────────────────────┘
                              │
┌──────────────────────────────────────────────────────────────┐
│                    CLOUD TIER (GCP)                          │
│                                                              │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────────┐   │
│  │ Vertex AI    │  │ BigQuery     │  │ Cloud Composer   │   │
│  │ - Training   │  │ - Analytics  │  │ - Pipeline       │   │
│  │ - Registry   │  │ - Events     │  │   Orchestration  │   │
│  │ - Monitoring │  │ - Compliance │  │ - Model Refresh  │   │
│  └──────────────┘  └──────────────┘  └──────────────────┘   │
└──────────────────────────────────────────────────────────────┘
                              │
┌──────────────────────────────────────────────────────────────┐
│                    EDGE TIER (Per Plant)                     │
│                                                              │
│  ┌──────────────────────────────────────────────────┐        │
│  │ NVIDIA GPU Edge Server (A2)                      │        │
│  │  ┌────────────┐  ┌────────────┐  ┌────────────┐  │        │
│  │  │ DeepStream │  │ Inference  │  │ Alert Engine │  │        │
│  │  │ Pipeline   │  │ Models     │  │ + Local    │  │        │
│  │  │ (Multi-    │  │ (16 Use    │  │   Buffer   │  │        │
│  │  │  Stream)   │  │  Cases)    │  │            │  │        │
│  │  └────────────┘  └────────────┘  └────────────┘  │        │
│  └──────────────────────────────────────────────────┘        │
│                              │                               │
│  ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐        │
│  │Camera 1  │ │Camera 2  │ │Camera 3  │ │Camera N  │        │
│  │(RTSP/    │ │(RTSP/    │ │(RTSP/    │ │(RTSP/    │        │
│  │ ONVIF)   │ │ ONVIF)   │ │ ONVIF)   │ │ ONVIF)   │        │
│  └──────────┘ └──────────┘ └──────────┘ └──────────┘        │
└──────────────────────────────────────────────────────────────┘
```

3.2 Edge Tier

Each plant receives one or more NVIDIA GPU edge servers running DeepStream pipelines for real-time inference:

- **Hardware**: NVIDIA A2 GPU in enterprise edge server (Dell PowerEdge / Advantech). Each server handles 38-45 simultaneous 1080p camera streams.
- **Software**: NVIDIA DeepStream SDK 7.x for multi-stream video pipeline management. Supports YOLO, SSD, FasterRCNN, and custom model architectures.
- **Camera Integration**: RTSP/ONVIF protocol support for existing camera infrastructure. No camera replacement required.
- **Local Processing**: All inference happens on-premises at the plant. Video data does not leave the plant network for real-time detection. Only metadata (detection events, timestamps, confidence scores) and short video clips (pre/post event) are sent to the cloud tier.
- **Alert Engine**: Local alert routing for sub-10-second notification delivery. Alert queue with retry logic for delivery assurance.
- **Local Buffer**: 24-72 hour video clip storage for event context (configurable based on storage capacity).

Sizing : ~45 edge servers for 1,969 cameras (A2-based). Additional servers provisioned for 5-10% camera growth.

3.3 Cloud Tier (GCP)

Centralized on Google Cloud Platform (Mumbai region for data residency):

- **Vertex AI**: Model training (AutoML for standard use cases, custom training for niche), model registry (version control, lineage), model monitoring (drift detection, accuracy tracking), edge model export (TensorRT optimized).
- **BigQuery**: Analytics data warehouse. All detection events, alert lifecycle data, compliance scores across all plants. Foundation for reporting and conversational analytics.
- **Cloud Composer**: Pipeline orchestration. Automated workflows for: event data → BigQuery ETL, annotated data → Vertex AI retraining, updated models → edge deployment.

3.4 Application Tier

- **Enterprise Dashboard (React)**: Real-time monitoring across all plants. Alert lifecycle management (new → acknowledged → resolved). Compliance dashboards (PPE scores, incident rates by plant/shift/zone). Use case performance monitoring (accuracy, false positive rates). Report generation and export.
- **Mobile App (React Native)**: Push notifications for critical alerts. Alert acknowledgment and resolution on-the-go. Key metrics and compliance scores. Plant-level drill-down. Available for iOS and Android.
- **Notification Engine**: Multi-channel alert delivery: SMS, WhatsApp, email. Configurable routing rules by use case, severity, plant, and role. Delivery confirmation and retry logic. Sub-10-second end-to-end delivery SLA.

3.5 Intelligence Tier (Conversational Analytics)

This tier goes beyond the RFP's dashboard requirement. Built on InsightLens's conversational analytics capability:

- **Natural Language Interface**: Powered by Google Gemini over BigQuery safety data. Plant managers and leadership query data conversationally instead of navigating predefined reports.
- **Prescriptive Recommendations**: The system doesn't just answer questions — it proactively surfaces insights and recommends actions based on pattern detection across plants.
- **Cross-Plant Benchmarking**: Compare safety metrics across all 44 plants. Identify best practices and lagging facilities.
- **Trend Analysis**: Automatic detection of improving or deteriorating safety trends.

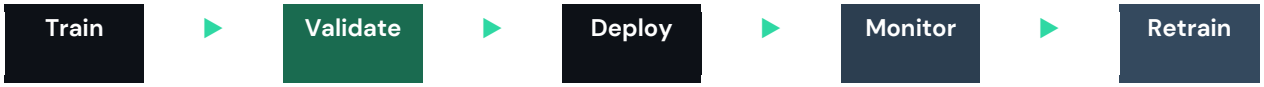
Example interactions:

- "Which plants have the highest PPE violation rate this month?" → Ranked list with comparison to prior month
- "What's driving the increase in fire alarms at Plant 12?" → Analysis showing weather-related false positives + recommendation to adjust threshold
- "How does our ANPR accuracy compare across entry points?" → Performance breakdown with recommended camera upgrades
- "Predict next month's safety incident rate based on current trends." → Statistical forecast with confidence interval

3.6 Data Flow

1. Camera streams (RTSP/ONVIF) → Edge GPU server
2. DeepStream pipeline → Multi-model inference (all applicable use cases per camera)
3. Detection event → Alert engine (immediate notification within 10s)
4. Detection metadata → BigQuery (batch ETL via Cloud Composer)
5. Video clips (event context) → Cloud Storage (configurable retention)
6. Annotated data (human-reviewed) → Vertex AI (periodic model retraining)
7. Updated models → Edge servers (OTA deployment via Vertex AI)
8. BigQuery data → Conversational Analytics UI (real-time querying)

3.7 Model Lifecycle



1. **Train**: Initial training on Vertex AI using Wobot's pre-trained models as starting point + UPL site-specific data
2. **Validate**: Automated accuracy testing against held-out test sets + human review
3. **Deploy**: TensorRT-optimized models deployed to edge servers via OTA update
4. **Monitor**: Continuous accuracy monitoring. Drift detection alerts when performance degrades.

5. **Retrain**: Triggered by accuracy drift or new annotated data. Automated pipeline via Cloud Composer.

4. Use Cases & Agentic Workflows

4.1 Use Case Detail

PPE Detection — Goggles

Attribute	Detail
Feasibility	Yes
Accuracy Target	90-95%
Detection Method	Multi-class object detection (YOLO v8/v9). Person detection → head ROI → goggles classification.
Technical Notes	Small object — requires adequate camera resolution (minimum 720p, recommended 1080p). Accuracy improves with closer camera mounting and consistent lighting.
Alert Workflow	Detection → classify severity (critical zone vs. general) → notify supervisor → dashboard update → compliance score update

PPE Detection — Safety Belt

Attribute	Detail
Feasibility	Yes
Accuracy Target	90-95%
Detection Method	Person detection → torso ROI → harness/belt classification. Pose estimation for at-height detection.
Technical Notes	Most relevant for elevated work areas. Camera angle matters — side/front views preferred over overhead.
Alert Workflow	Detection → check zone (elevated work zone) → immediate alert to safety officer → dashboard update

PPE Detection — Suit

Attribute	Detail
Feasibility	Yes
Accuracy Target	90-95%
Detection Method	Full-body object detection. Classify suit type (chemical protection, general workwear) based on color/pattern.
Technical Notes	Easier than goggles (larger object). Color-coded suits simplify detection. Site-specific training for UPL's suit types.
Alert Workflow	Detection → zone-based enforcement (chemical handling requires specific suit) → alert → compliance score

PPE Detection — Shoes

Attribute	Detail
Feasibility	Conditional

Accuracy Target	88-93%
Detection Method	Person detection → foot ROI → safety footwear classification.
Technical Notes	Very small object in typical plant camera frames. Accuracy depends heavily on camera angle and mounting height. Ground-level or low-angle cameras significantly improve detection. May require additional cameras in critical zones.
Alert Workflow	Detection → alert if in designated zone → dashboard update

Man Down

Attribute	Detail
Feasibility	Yes
Accuracy Target	90-95%
Detection Method	Pose estimation (body keypoints) + temporal behavioral analysis. Detect standing/walking → prone transition. Configurable dwell-time threshold (15-30s).
Technical Notes	Must distinguish collapse from crouching/bending. Dwell-time threshold critical for false positive management. Works well with overhead and angled cameras.
Alert Workflow	Detection → dwell-time confirmation → EMERGENCY alert to safety officer + plant manager → automatic location relay (camera ID, zone) → dashboard critical alert

Tanks Overfill/Vent

Attribute	Detail
Feasibility	Conditional
Accuracy Target	88-93%
Detection Method	Anomaly detection + semantic segmentation. Baseline normal tank state, detect deviations. Liquid level estimation via visual markers. Vent detection via visible emission patterns.
Technical Notes	Niche use case. Requires significant custom training with UPL site-specific data (each tank type, each chemical). Recommend dedicated cameras with consistent angle on critical tanks. Hybrid physics-informed + data-driven models for best accuracy.
Alert Workflow	Detection → classify severity (overfill vs. vent) → immediate alert to process operator → dashboard critical alert

Machine Guarding

Attribute	Detail
Feasibility	Yes
Accuracy Target	90-95%
Detection Method	Zone-based intrusion detection. Define exclusion

	zones around active machinery. Detect human presence within zone.
Technical Notes	Requires ROI mapping per machine per camera. Must integrate with machine operational status (running/stopped) to avoid false alerts during maintenance. Person detection + zone intersection calculation.
Alert Workflow	Detection → check machine status (running?) → if running, immediate alert to floor supervisor → dashboard alert

Overcrowding

Attribute	Detail
Feasibility	Yes
Accuracy Target	92-95%
Detection Method	Density estimation (CSRNet/MCNN) or person detection + counting in defined zones. Configurable maximum occupancy per zone.
Technical Notes	Works well with fixed overhead cameras. Calibration needed per camera for pixel-to-person mapping. Zone definitions customizable per plant layout.
Alert Workflow	Detection → compare to zone threshold → warning at 80%, alert at 100% → notify zone supervisor → dashboard update

Over Speeding

Attribute	Detail
Feasibility	Conditional
Accuracy Target	90-95%
Detection Method	Vehicle detection + speed estimation via calibrated cameras. Pixel displacement across known real-world distances between frames.
Technical Notes	Requires camera calibration (focal length, mounting angle, known reference distances). Frame rate matters — minimum 15fps recommended. Dual-camera setups improve accuracy. Speed threshold configurable per zone (loading area, main road, warehouse).
Alert Workflow	Detection → compare to zone speed limit → alert with vehicle image + estimated speed → dashboard log

Using Fire Hydrant Hose

Attribute	Detail
Feasibility	Conditional
Accuracy Target	88-93%
Detection Method	Object state detection. Classify fire hydrant hose as "deployed" vs. "stored" state.
Technical Notes	Niche use case. Requires custom training with UPL site-specific examples. Binary classification is

	relatively straightforward once training data is sufficient. May need to distinguish authorized (fire drill) from unauthorized use via schedule integration.
Alert Workflow	Detection → check schedule (drill/maintenance?) → if unscheduled, alert to safety + security → dashboard log

Dust Cloud

Attribute	Detail
Feasibility	Conditional
Accuracy Target	85-92%
Detection Method	Anomaly detection with temporal analysis. Similar to smoke detection but with distinct visual signatures. Environmental context modeling to distinguish normal dust (grinding, loading) from abnormal concentrations.
Technical Notes	High environmental variance. Wind, humidity, and normal operations generate dust. False positive management critical. May benefit from multi-sensor fusion (air quality sensors + visual).
Alert Workflow	Detection → severity classification (minor/major) → alert to EHS team → dashboard update

Fire & Smoke

Attribute	Detail
Feasibility	Yes
Accuracy Target	95-97%
Detection Method	Dual-model: (1) CNN/YOLO for visual flame and smoke appearance, (2) temporal analysis for smoke behavior (rising, expansion rate).
Technical Notes	Well-studied use case with high accuracy benchmarks (91-98.7%). False alarm rate achievable below 2%. Multi-spectral camera recommendation for chemical areas. Integrates with existing fire alarm systems for corroboration.
Alert Workflow	Detection → EMERGENCY classification → immediate alert to fire safety team + plant manager → dashboard critical alert → automatic video clip capture (pre + post event)

ANPR (Automatic Number Plate Recognition)

Attribute	Detail
Feasibility	Yes
Accuracy Target	95-98%
Detection Method	Two-stage: (1) plate localization via object detection, (2) character recognition via OCR trained on Indian plate formats.
Technical Notes	Proven technology in India (FASTag, Smart Cities Mission). Requires dedicated ANPR cameras at entry/exit points (high resolution, IR illumination for night). Multiple Indian plate format support (state

	codes, commercial/private).
Alert Workflow	Detection → match against authorized vehicle list → if unauthorized, alert to security → log all entries/exits → dashboard vehicle management view

Mobile Phone Usage

Attribute	Detail
Feasibility	Yes
Accuracy Target	92-95%
Detection Method	Person detection → hand/head ROI → phone object detection + pose estimation (arm position indicating phone use).
Technical Notes	Small object challenge. Two-step detection reduces false positives. Higher accuracy with closer camera mounting. May confuse radio usage — configurable sensitivity. Zone-based enforcement (hazardous zones only).
Alert Workflow	Detection → zone check (hazardous area?) → alert to supervisor → dashboard log

Low Lux

Attribute	Detail
Feasibility	Yes (camera capability, not separate model)
Accuracy Target	85-92% (model performance in low light)
Detection Method	Not a separate detection model. All models must perform in low-light conditions. Addressed through camera hardware (IR/thermal) and training data augmentation.
Technical Notes	IR-enabled cameras for areas with reduced lighting. Thermal cameras for pitch-dark areas or areas where visual cameras cannot perform. Training data must include low-light augmented images. Performance degradation without proper cameras: 10-15%.
Recommendation	Identify all low-lux zones during Phase 1 (Discovery). Specify IR-enabled or thermal cameras for UPL procurement. Include low-light test scenarios in UAT.

Spillage

Attribute	Detail
Feasibility	Yes
Accuracy Target	90-95%
Detection Method	Semantic segmentation to detect liquid on surfaces where it shouldn't be. Temporal anomaly detection for sudden appearance of liquid. Color and texture analysis for chemical identification.
Technical Notes	Different chemicals have different visual signatures. Requires site-specific training with UPL's actual chemical types. Wet floor vs. spillage distinction needed. Environmental factors (rain, cleaning) managed via schedule integration.

Alert Workflow	Detection → chemical identification attempt → alert to EHS team + response team → dashboard critical alert → containment protocol triggered

4.2 Agentic Workflow: Detection-to-Resolution

Every detection event follows a standard agentic workflow:

1. **Detection**: AI model identifies event in video stream
2. **Classification**: Event classified by use case, severity (critical/warning/info), zone, shift
3. **Routing**: Alert routed to appropriate personnel based on configurable rules (use case + severity + zone → responsible person)
4. **Notification**: Multi-channel delivery (SMS, WhatsApp, email) within 10-second SLA
5. **Acknowledgment**: Operator acknowledges alert in dashboard or mobile app (timestamp recorded)
6. **Resolution**: Operator records resolution action, attaches evidence if needed, closes alert
7. **Analytics**: Event data flows to BigQuery. Conversational analytics surfaces patterns and prescriptive recommendations.

Agentic Enhancement : An intelligent alert tuning agent monitors false positive patterns and recommends threshold adjustments. An automated retraining agent triggers model updates when accuracy drift is detected. A prescriptive analytics agent surfaces cross-plant patterns and recommends interventions.

5. Delivery Approach

5.1 Phased Delivery

We propose a compressed pilot-to-rollout approach that delivers full production capability within 12-14 weeks while de-risking through a structured pilot phase.

Phase 1: Discovery & Setup (Weeks 1-2)

Activity	Deliverable
Camera infrastructure survey (remote + on-site)	Camera inventory with specs, quality assessment, use case mapping
Network assessment	Bandwidth and connectivity analysis per plant
ROI definition: use case → camera → zone mapping	ROI configuration document
Hardware specification and ordering	Purchase orders for edge servers
GCP environment setup	Vertex AI project, BigQuery dataset, Cloud Composer environment
Data collection kick-off	Initial video samples for model training

Gate Criteria : Camera survey complete, hardware ordered, GCP environment ready, ROI mapping approved by UPL.

Phase 2: Pilot Build & Deploy (Weeks 3-8)

Activity	Deliverable
Model development for 5 priority use cases	Trained models (PPE, fire/smoke, man down, ANPR, overcrowding)
Edge deployment at 1-2 pilot plants	Working inference pipeline on edge servers
Dashboard MVP development	Web dashboard with real-time monitoring, alert lifecycle
Mobile app v1 development	React Native app with push notifications, alert acknowledgment
Notification engine integration	SMS, WhatsApp, email delivery working within SLA
Integration testing	End-to-end testing across all components
UAT at pilot plants	User acceptance testing with plant safety teams
Accuracy validation	Accuracy report per use case at pilot plants
Conversational analytics MVP	Basic natural language querying over pilot data

Gate Criteria : 5 use cases operational at pilot plants, accuracy targets met, dashboard + mobile app functional, UAT signed off.

Build → Test → Deploy Pattern :

- Weeks 3-5: Build (model dev, dashboard, mobile, integration)
- Weeks 6-7: Test (integration testing, QA, performance testing, security testing)
- Week 8: Deploy (pilot plant go-live, UAT, accuracy validation)

Phase 3: Rollout (Weeks 9-13)

Activity	Deliverable
Remaining 11 use case model development	Trained models for all 16 use cases
Edge server deployment to remaining plants	Infrastructure operational across all sites

Model replication to all plants	All use cases running at all plants
Full dashboard + conversational analytics	Complete enterprise dashboard with conversational AI
Full mobile app release	Production mobile app with all features
Cross-plant testing	Verification across all deployments
Training for UPL operations teams	Trained operators at each plant
Performance validation	Accuracy report across all plants and use cases

Gate Criteria : All 16 use cases operational across all plants, accuracy targets met, all plant operations teams trained, full dashboard + conversational analytics operational.

Build → Test → Deploy Pattern :

- Weeks 9-10: Build (remaining models, full dashboard, conversational UI)
- Weeks 11-12: Test (cross-plant testing, regression, performance, security)
- Week 13: Deploy (full rollout, training, validation)

Phase 4: Operate & Optimize (Ongoing)

Activity	Deliverable
Managed service operation	24/7 system monitoring, SLA compliance
Continuous model improvement	Accuracy improvements based on production data
New use case development	Additional use cases as UPL identifies needs
Model retraining pipeline	Automated retraining triggered by drift detection
Quarterly business reviews	Performance reports, ROI analysis, roadmap alignment

5.2 Sprint Cadence

Throughout all phases:

- **Daily**: Stand-up with tvameva delivery team
- **Weekly**: Progress review with UPL project team
- **Bi-weekly**: Steering committee with UPL leadership
- **Per Phase**: Gate review and go/no-go decision

6. Team Composition

6.1 AI-Enabled Pod

Role	Count	Responsibility	Source
Solution Architect	1	End-to-end architecture, integration design, stakeholder alignment	tvameva
CV/ML Engineer	2	Model development, fine-tuning, accuracy optimization, TAO Toolkit	tvameva + Wobot
Platform Engineer (GCP)	1	Vertex AI pipelines, BigQuery, Cloud Composer, infrastructure	tvameva
Full-Stack Developer	1	Enterprise dashboard (React), notification engine	tvameva
Mobile App Developer	1	React Native mobile app, push notifications	tvameva
QA Engineer	1	Model validation, integration testing, UAT coordination	tvameva
Hardware & Deployment Engineer	1	Edge server procurement, deployment, camera integration	Hardware Partner
Customer Success Lead	1	Checkpoint facilitation, training, ongoing relationship	tvameva
Total	**9**		

6.2 Agentic Augmentation

In addition to the human team, agentic workflows handle:

Agentic Workflow	Function	Impact
Automated Model Retraining	Monitor accuracy drift, trigger retraining pipelines, validate updated models	Continuous improvement without manual intervention
Intelligent Alert Tuning	Learn false positive patterns, recommend threshold adjustments per plant/camera	Reduced alert fatigue, improved signal-to-noise ratio
Agentic QA	Automatically test model performance across edge conditions (lighting, weather, angles)	Broader test coverage, faster validation
Deployment Orchestration	Automated model deployment to edge servers, rollback on failure	Consistent deployments across 44 plants
Prescriptive Analytics Agent	Surface cross-plant patterns, generate safety recommendations	Proactive safety management, not reactive

6.3 Technology Partners

Partner	Role	Credentials
Wobot Intelligence	CV model development, edge deployment expertise	India-based (New Delhi), NVIDIA technology partner, 99.2% accuracy on manufacturing CV, clients include IRCTC/Indian Railway, backed by Peak XV Partners
Dell / Advantech	GPU edge server procurement, on-site deployment	NVIDIA Metropolis system builder partners, India presence, enterprise hardware support

7. Pricing & Commercial Model

tvameva provides three commercial models per the RFP structure. All amounts in INR.

Option A: One-Time Development + Annual AMC (CAPEX Model)

Particulars	Amount (INR Cr)
a. One-Time Cost — Development	
AI/ML Software Development (16 use cases)	4.50
Enterprise Dashboard + Mobile App	1.20
Conversational Analytics UI	0.80
System Integration & Camera Mapping	1.20
Notification Engine	0.30
Training & Change Management	0.50
Subtotal: Development	**8.50**
b. Recurring Cost / AMC (Yearly)	
Software Maintenance & Support	0.85
Model Retraining & Optimization	0.30
Conversational Analytics Operations	0.15
Subtotal: Annual AMC	**1.30**
c. Hardware Cost (Edge Computing Servers with GPU)	
~45 NVIDIA A2 Edge Servers (Dell/Advantech)	5.00
Network Equipment & Accessories	0.50
Subtotal: Hardware	**5.50**
d. New Requirement Cost (per additional model/dashboard)	0.15 - 0.30 per use case
e. Cloud Hosting Cost (GCP — Vertex AI, BigQuery, Storage)	0.50 /year
f. Installation & Commissioning	0.50
Total One-Time Investment (a + c + f)	**14.50**
Total Annual Recurring (b + e)	**1.80**
5-Year TCO	**23.50**

Option B: SaaS Model (Per-Camera/Month)

Particulars	Amount (INR)
Per-Camera Per-Month Fee (all use cases included)	3,750
Monthly Total (1,969 cameras)	73.84 Lakhs
Annual SaaS Fee	**8.86 Cr**
One-Time Setup:	
Hardware (edge servers, UPL-owned)	5.50 Cr
Installation & Commissioning	0.50 Cr

Initial Integration & Camera Mapping	1.50 Cr
Total One-Time	**7.50 Cr**
Cloud hosting, AMC, model retraining, support included in SaaS fee	
5-Year TCO	**51.80 Cr**

Option C: Perpetual License

Particulars	Amount (INR Cr)
g. Perpetual License (Software — all use cases)	8.00
c. Hardware Cost (Edge Servers with GPU)	5.50
f. Installation & Commissioning	0.50
Total One-Time (g + c + f)	**14.00**
b. Annual AMC (20% of license)	1.60
e. Cloud Hosting (annual)	0.50
d. New Requirement Cost (per additional model)	0.15 - 0.30
Total Annual Recurring (b + e)	**2.10**
5-Year TCO	**24.40**

Payment Terms

CAPEX / Perpetual License Models:

- 30% on contract signing
- 30% on pilot completion (Phase 2 gate)
- 30% on full rollout completion (Phase 3 gate)
- 10% on UAT sign-off and go-live

SaaS Model:

- One-time fees: 50% on signing, 50% on deployment
- SaaS fees: Monthly invoicing, 30-day payment terms

Recommendation

We recommend Option A (One-Time + AMC) for UPL. It provides the lowest 5-year TCO (INR 23.50 Cr vs. 51.80 Cr for SaaS), gives UPL full ownership of the deployed software, and aligns with UPL's CAPEX budgeting for manufacturing infrastructure investments. The annual AMC ensures continuous model improvement and support.

8. Timeline & Milestones

8.1 Project Timeline

```
Week:  1  2  3  4  5  6  7  8  9  10  11  12  13  14
       ├──┼──┼──┼──┼──┼──┼──┼──┼──┼───┼───┼───┼───┤
Phase: │P1│  │      Phase 2       │     Phase 3      │
       │  │  │  Pilot Build &     │     Rollout       │
       │DI│  │  Deploy            │                   │
       │SC│  │                    │                   │
       │  │  │                    │                   │
       └──┘  └────────────────────┴───────────────────┘
```

8.2 Key Milestones

Milestone	Week	Acceptance Criteria
M1: Discovery Complete	2	Camera survey done, ROI mapping approved, hardware ordered, GCP ready
M2: Pilot Models Ready	5	5 priority use cases trained, accuracy validated on test data
M3: Pilot Dashboard MVP	6	Dashboard functional with real-time monitoring and alert lifecycle
M4: Pilot Go-Live	8	1-2 plants operational, accuracy targets met, UAT signed off
M5: All Use Cases Ready	10	16 use cases trained and validated
M6: Full Dashboard + Conversational Analytics	11	Complete enterprise dashboard with conversational AI operational
M7: Full Rollout Complete	13	All plants operational, all use cases running
M8: Production Sign-Off	14	Full system validated, teams trained, SLA monitoring active

8.3 Critical Path Items

1. **Hardware procurement lead time**: GPU edge servers must be ordered in Week 1 for delivery by Week 3-4. Any delay in procurement directly impacts the timeline.
2. **UPL camera data access**: Training data collection begins Week 1. Delays in site access or video feed provisioning affect model development start.
3. **Site access for deployment**: On-site installation windows must be coordinated with plant operations schedules.

9. Risk & Mitigation

9.1 Decision & Strategic Risks

Risk	Impact	Probability	Mitigation
Hardware procurement delays	Timeline slip. Edge servers not available for pilot.	Medium	Order in Week 1. Identify backup hardware vendor. Pre-stage servers for pilot plants.
Camera infrastructure gaps	Some use cases infeasible at certain plants due to camera quality, positioning, or coverage.	Medium	Phase 1 camera survey identifies gaps early. UPL procures additional cameras per our specification. Pilot plants selected for optimal camera infrastructure.
Site access coordination	Deployment delayed if plant access windows are constrained by production schedules.	Medium	Agree access schedule during Phase 1. Weekend/shutdown windows for installation. Remote pre-configuration to minimize on-site time.
Network bandwidth at remote plants	Edge-to-cloud communication unreliable, affecting model updates and analytics.	Low-Medium	Solution designed for edge-first operation. Plants function independently if cloud connectivity is intermittent. Cloud sync is batch, not real-time dependency.

9.2 Technical Risks

Risk	Impact	Probability	Mitigation
Niche use case accuracy (tanks overfill, dust cloud, fire hydrant hose)	Accuracy below 90% initially for niche use cases due to limited pre-trained models.	Medium	Transparent accuracy commitments per use case (see Appendix A). Iterative fine-tuning in Phase 4. Dedicated camera recommendations for niche use cases. UPL acceptance that niche use cases reach target accuracy over 2-3 iteration cycles, not at initial deployment.
False positive fatigue	High false positive rates on certain use cases reduce operator trust and engagement.	Medium	Intelligent alert tuning agent learns patterns and adjusts thresholds. Severity-based routing (only critical alerts notify via SMS/WhatsApp).

			Dashboard shows confidence scores for operator judgment. Monthly false positive analysis and tuning.
Low-lux performance	Accuracy degrades significantly in low-light areas without proper cameras.	Low	IR-enabled and thermal camera recommendations provided in Phase 1. Low-light areas identified and flagged. All models trained with low-light augmented data.
Model drift over time	Accuracy degrades as environmental conditions change (seasons, new equipment, camera repositioning).	Medium	Automated drift detection via Vertex AI model monitoring. Retraining pipeline triggered automatically. Monthly accuracy reports and trend analysis.

9.3 Data & Compliance Risks

Risk	Impact	Probability	Mitigation
Data residency requirements	UPL may require all data to remain in India. Certain GCP services may have region limitations.	Low	Default to GCP Mumbai region (asia-south1). Validate all services available in-region during Phase 1. Edge processing keeps video data on-premises.
Privacy concerns with video surveillance	Regulatory or employee relations issues around AI-powered surveillance.	Low-Medium	System focuses on safety events, not employee tracking. No facial recognition. No individual performance monitoring. UPL to manage employee communication and consent per local regulations. tvameva provides privacy impact assessment template.

10. Terms & Conditions

10.1 Scope

This proposal covers the design, development, deployment, and initial operation of an AI-based video analytics system for plant safety and security as described in Sections 1-9. Any scope changes require mutual written agreement and may affect timeline and pricing.

10.2 Intellectual Property

- **UPL Data Ownership**: UPL retains full ownership of all video data, annotations, processed outputs, and derivative insights generated by the system. Per Annexure-1 requirements.
- **Model Artifacts**: Models trained specifically for UPL using UPL's data are UPL's intellectual property. Model weights and artifacts will not be reused for other customers without UPL's written consent.
- **Platform IP**: tvameva's platform components (dashboard framework, notification engine, deployment orchestration) remain tvameva's intellectual property, licensed to UPL for the duration of the engagement.
- **Partner IP**: Wobot Intelligence's pre-trained base models remain Wobot's IP, licensed to UPL through the engagement. UPL-specific fine-tuned layers are UPL's IP.

10.3 Confidentiality

All information exchanged between parties is treated as confidential. Neither party will disclose the other's confidential information to third parties without written consent. Standard NDA terms apply.

10.4 Service Level Agreements

Per RFP requirements, tvameva commits to:

Service	SLA	Measurement
Remote Support	Resolution within 4 hours	Ticket creation to resolution
On-Site Support	Resolution within 24-48 hours	For critical codebase/model issues
System Uptime	99.5%	Monthly measurement, excluding planned maintenance
Alert Delivery	Within 10 seconds	Detection to notification delivery
Model Fine-Tuning	Within 1 week	Request to deployed improvement

SLA measurement begins after Phase 3 (Rollout) completion.

10.5 Exit Clause

Per RFP terms, UPL has the right to terminate services with 1 month prior written notice. Upon termination:

- All UPL data and model artifacts will be transferred to UPL within 15 business days
- tvameva will provide transition support documentation
- Hardware owned by UPL remains with UPL
- Outstanding invoices for completed work are payable within 30 days

10.6 Warranty

- 90-day warranty on all software deliverables from Phase 3 completion
- During warranty period, bug fixes and critical issues resolved at no additional cost
- Warranty does not cover new features, new use cases, or changes to scope

10.7 Limitation of Liability

tvameva's total liability is limited to the total fees paid under the engagement. Neither party is liable for indirect, consequential, or incidental damages except in cases of willful misconduct or gross negligence.

11. Cybersecurity & Data Ownership Response

Detailed response to RFP Annexure-1, all 13 sections.

11.1 Data Security & AI Governance Framework

Security Frameworks:

- **ISO 27001**: Information Security Management System. tvameva follows ISO 27001 controls for all development and operational practices. GCP infrastructure is ISO 27001 certified.
- **NIST Cybersecurity Framework (CSF)**: Our security practices align with NIST CSF functions: Identify, Protect, Detect, Respond, Recover.

AI Governance Frameworks:

- **ISO/IEC 42001**: AI Management Systems. tvameva is adopting ISO 42001 practices for responsible AI governance, including model risk management, bias assessment, and transparency.
- **NIST AI RMF**: Risk Management Framework for AI. Applied to all model development — risk identification, measurement, and management throughout the AI lifecycle.

Documentation Available:

- AI governance policy document
- Model risk register template (populated per use case during development)
- AI transparency documentation: model cards for each production model (purpose, training data description, known limitations, evaluation metrics, risk assessment)
- Data sheets for training datasets (sources, composition, preprocessing, known biases)

11.2 Data & Model Ownership / Usage

Ownership Confirmation:

- UPL retains **full ownership** of all video/AI data: raw video, processed images, annotations, metadata, model outputs, and any derivative insights.
- UPL retains ownership of model artifacts (weights, checkpoints, embeddings) developed using UPL's data.

Prohibited Usage:

- UPL's data will **NOT** be used to train, fine-tune, or improve models for any other customer.
- UPL's data will **NOT** be used to train or improve any general-purpose or foundation models.
- Model artifacts developed for UPL will **NOT** be reused, provided to, or leveraged for any other customer without UPL's prior written consent.

Technical Controls:

- **Tenant Isolation**: UPL's GCP project is fully isolated. Separate project, separate service accounts, separate data buckets, separate Vertex AI endpoints.
- **Dataset Controls**: Training pipelines configured with explicit dataset allow-lists. Only UPL-tagged datasets can feed UPL model training jobs.

- **Segregated Storage**: All UPL data in dedicated Cloud Storage buckets with IAM policies restricting access to UPL project service accounts only.
- **DLP Rules**: Google Cloud DLP configured to detect and prevent data leakage across project boundaries.

Verification:

- **Data lineage reports**: Available via Vertex AI — every training job records dataset versions, preprocessing steps, and output model artifacts.
- **Model registry entries**: Every model version records training data source, hyperparameters, evaluation metrics, and deployment history.
- **Training job attestations**: Automated logs of every training run with dataset references and compute resources used.
- **Access logs**: Cloud Audit Logs for all data access, model registry events, and administrative actions.
- **Audit rights**: UPL has the right to audit all controls, logs, and practices upon reasonable notice.

11.3 Data Encryption

At Rest:

- AES-256 encryption for all data stored in GCP (Cloud Storage, BigQuery, Vertex AI artifacts)
- Google-managed encryption keys by default, with option for Customer-Managed Encryption Keys (CMEK) via Cloud KMS
- Model artifacts (checkpoints, embeddings, registry entries) encrypted at rest

In Transit:

- TLS 1.2+ for all communications between edge servers and cloud
- mTLS available for service-to-service communication
- All API endpoints enforce HTTPS

During Processing:

- GCP Confidential Computing available for sensitive workloads (Confidential VMs with AMD SEV or Intel TDX)
- GPU memory isolation during model training and inference

Key Management:

- Google Cloud KMS for key management
- Key rotation: automatic rotation every 90 days (configurable)
- Per-tenant key segregation: UPL's encryption keys are isolated from all other projects
- HSM-backed keys available via Cloud HSM for highest assurance

11.4 Data Residency

Storage & Processing Locations:

- **All data**: Stored and processed in GCP Mumbai region (asia-south1) by default

- **Live video**: Processed on-premises at edge servers. Video does not leave UPL plant network for real-time inference.
- **Detection metadata**: Stored in BigQuery (asia-south1)
- **Model artifacts**: Stored in Vertex AI model registry (asia-south1)
- **Training**: Conducted on Vertex AI in asia-south1 region
- **Analytics**: BigQuery processing in asia-south1

Residency Controls:

- UPL can pin data residency to India (asia-south1). No data will be stored or processed outside India without UPL's written approval.
- Cross-border transfer controls: Organization policies in GCP enforcing region restriction. No data export without explicit policy change approved by both parties.
- Data minimization: Only detection metadata and event clips are sent to cloud. Raw video streams remain on-premises.

11.5 Access Control

Identity & Access:

- **RBAC**: Role-based access control via GCP IAM. Predefined roles: Admin, Developer, Operator, Viewer, Auditor.
- **SSO/SAML/OIDC**: Integration with UPL's identity provider for single sign-on.
- **MFA**: Multi-factor authentication required for all human access to GCP resources and dashboard.
- **JIT Access**: Just-in-time privileged access for administrative operations. Elevated privileges granted for specific duration with approval workflow.
- **Break-Glass**: Emergency access procedure for critical incidents. Logged, time-limited, reviewed post-incident.
- **Quarterly Reviews**: Access reviews conducted quarterly. Stale accounts deactivated. Excessive permissions flagged and remediated.

Environment Isolation:

- Separate GCP projects for development, staging, and production
- No cross-environment data access
- Model promotion requires explicit approval workflow: dev → staging (automated tests) → production (human approval)

Model Promotion Controls:

- Development models cannot be deployed to production without passing automated accuracy tests
- Staging validation against holdout test set required before production promotion
- Production deployments logged with rollback capability

Secrets Management:

- Google Secret Manager for all credentials, API keys, and sensitive configuration

- Automatic rotation on configurable schedules
- No secrets stored in code, configuration files, or environment variables in images

11.6 Incident Response

IR Plan:

- tvameva maintains an Incident Response Plan covering identification, containment, eradication, recovery, and lessons learned.
- Designated IR contact provided to UPL at engagement start.

Notification SLAs:

- **Initial notification**: Within 24 hours of confirmed security incident
- **Containment update**: Within 48 hours with initial impact assessment
- **Root cause analysis**: Within 5 business days
- **Full incident report**: Within 10 business days

AI-Specific Incidents:

- **Data leakage / model inversion**: Detection via access logs and output analysis. Containment via immediate model endpoint shutdown.
- **Data/model poisoning**: Detection via model monitoring (accuracy drift, anomalous training data). Containment via model rollback to last known good version.
- **Prompt/data injection**: Input validation on all API endpoints. Anomalous input detection and logging.
- **Compromised model registry**: Integrity verification on all model artifacts. Signed model checksums.
- **Drift causing harmful outcomes**: Continuous model monitoring. Automatic alerts on accuracy degradation. Human review before retraining deployment.

Communication Artifacts:

- Incident timeline with UTC timestamps
- Impact assessment (data affected, systems impacted, users affected)
- Evidence preservation (logs, artifacts)
- Remediation actions taken
- Lessons learned and preventive measures

11.7 Third-Party Risk

Sub-Processors / Providers:

Provider	Service	Location	Role
Google Cloud Platform	Cloud infrastructure, Vertex AI, BigQuery	Mumbai (asia-south1)	Cloud compute, storage, ML platform
Wobot Intelligence	CV model development, edge expertise	New Delhi, India	Pre-trained models, edge deployment
Dell / Advantech	Edge server hardware	India (procurement)	GPU server manufacturing and supply

NVIDIA	GPU hardware, DeepStream SDK, TAO Toolkit	Global (software), India (hardware via partners)	Edge inference platform

Flow-Down Obligations:

- All sub-processors contractually bound to UPL's data/model usage restrictions
- DPAs (Data Processing Agreements) in place with all sub-processors
- Sub-processors cannot access UPL data without explicit project-level IAM authorization
- Change notification: UPL will be notified 30 days in advance of any sub-processor change

Due Diligence:

- GCP: SOC 1/2/3, ISO 27001/27017/27018, PCI DSS certified. Audit reports available.
- Wobot: Security review conducted by tvameva. NDA and DPA in place.
- Annual security review of all sub-processors
- Penetration testing conducted annually on deployed systems

11.8 Data Retention & Deletion

Retention Defaults:

Data Type	Default Retention	Configurable
Raw video (edge)	24-72 hours (local buffer)	Yes
Event video clips (cloud)	90 days	Yes
Detection metadata	2 years	Yes
Annotations	Duration of engagement	Yes
Model artifacts	Duration of engagement + 1 year	Yes
Audit logs	3 years	Per UPL policy
Training data	Duration of engagement	Yes

Deletion:

- UPL can request secure deletion at any time, including backups and replicas
- **Method**: Cryptographic erasure (crypto-shredding) for cloud data. Secure overwrite for local storage.
- **Verification**: Certificate of destruction provided within 15 business days of deletion request
- **Scope**: Deletion covers all copies — primary storage, backups, replicas, and cached data

11.9 Vulnerability Management

Scanning & Patching:

- Automated vulnerability scanning: OS, containers, CUDA drivers, Python dependencies, ML pipeline components
- Container image scanning on every build (GCP Container Analysis)
- Dependency scanning via automated tools integrated into CI/CD

Patching Commitments:

- **Critical/High vulnerabilities**: Mitigated within 15 days of disclosure
- **Medium/Low vulnerabilities**: Mitigated within 30 days of disclosure
- **Evidence**: Ticketing system reports available to UPL upon request. Vulnerability tracking dashboard access for UPL security team.

ML-Specific:

- Model dependency scanning (TensorFlow, PyTorch, NVIDIA libraries)
- DeepStream SDK updates tracked and applied per NVIDIA release schedule
- TAO Toolkit and NIM microservice updates validated in staging before production

11.10 Regulatory Compliance

Applicable Regulations:

- **India DPDP Act 2023**: Digital Personal Data Protection. Consent management, data localization (India), breach notification (72 hours), data principal rights.
- **GDPR**: If UPL data subjects include EU residents. Data minimization, purpose limitation, right to erasure.
- **CCPA**: If applicable to UPL's California operations.
- **Sectoral**: Chemical industry safety regulations (OSHA equivalents in India — Factories Act, DISH rules).

Compliance Measures:

- DPIAs (Data Protection Impact Assessments) conducted for video analytics processing
- Records of processing activities maintained
- Consent/legal basis documented (legitimate interest for workplace safety)
- Export controls: No data or model artifacts exported to restricted jurisdictions
- Video analytics focused on safety events (PPE, hazards), not facial recognition or individual tracking — minimizes privacy regulatory exposure

11.11 Audit, Logging & Observability

What is Logged:

Category	Events Logged
Data access	All reads/writes to storage, BigQuery queries, model training data access
Authentication	Login attempts, SSO events, MFA challenges, session management
Authorization	Permission changes, role assignments, IAM policy updates
Configuration	Infrastructure changes, service configuration, deployment updates
Dataset & model registry	Dataset version creation, model registration, annotation updates
Training runs	Job start/end, dataset references, hyperparameters, compute resources, metrics
Model promotions	Dev → staging → production transitions, approval records
Inference	Metadata only (camera ID, use case, confidence,

	timestamp). NOT raw video.
Drift/quality monitors	Accuracy measurements, drift alerts, threshold breaches
Administrative	User management, project settings, billing events

Retention:

- Audit logs: 3 years (configurable per UPL policy)
- Operational logs: 90 days (configurable)
- All logs stored in GCP Cloud Logging (asia-south1)

Access & Export:

- UPL can access all relevant logs via Cloud Logging console or API
- SIEM integration: Logs exportable to UPL's SIEM via Cloud Logging export sinks (Pub/Sub, Cloud Storage, BigQuery)
- Log exports within 5 business days upon request

MLOps Tooling:

- **Vertex AI Model Registry**: Model versions, training datasets, evaluation metrics, deployment history
- **Vertex AI Experiments**: Training run tracking, hyperparameter comparison
- **Vertex AI Pipelines**: Pipeline execution logs, data lineage
- Role-based access to MLOps tools: Data Scientists (full), Engineers (deploy), UPL Admins (view all), UPL Auditors (read-only)
- All artifacts versioned and signed (model checksums, dataset hashes)

Dashboards & Alerts:

- Operational dashboard: throughput (inferences/second), inference errors, model latency
- Quality dashboard: accuracy per use case, drift trends, false positive rates
- Infrastructure dashboard: GPU utilization, edge server health, network connectivity
- Resource dashboard: GCP costs, storage usage, compute utilization
- Configurable alerts for threshold breaches on all metrics

11.12 Business Continuity & Disaster Recovery

Edge Tier (Per Plant):

- Edge servers operate independently. If cloud connectivity is lost, edge inference continues uninterrupted.
- Local alert delivery (SMS, WhatsApp, email) continues via direct connectivity.
- Local video buffer provides 24-72 hours of event storage.
- When connectivity restores, metadata automatically syncs to cloud.

Cloud Tier:

- GCP provides multi-zone availability within asia-south1 region.
- BigQuery: automatic replication across zones. 99.99% SLA.

- Vertex AI: model registry and endpoints available across zones.
- Cloud Storage: multi-zone redundancy by default.

RPO/RTO Targets:

- **Edge (critical)**: RPO = 0 (no data loss — edge operates independently), RTO = 0 (no downtime — edge continues operating)
- **Cloud analytics**: RPO = 1 hour (batch sync interval), RTO = 4 hours (GCP infrastructure recovery)
- **Dashboard/App**: RPO = 0 (stateless application), RTO = 1 hour (re-deployment from container registry)

DR Testing:

- Quarterly DR test: simulate cloud connectivity loss, verify edge-first operation
- Annual full DR exercise: simulate regional outage, verify recovery procedures
- DR test results shared with UPL

11.13 AI Logging & Transparency

Model Cards:

Each production model includes a model card documenting:

- Purpose and intended use
- Training data sources and composition (no PII in descriptions)
- Model architecture and version
- Evaluation metrics (accuracy, precision, recall, F1, false positive rate)
- Known limitations and failure modes
- Risk assessment per use case
- Recommended deployment conditions (camera requirements, lighting, angle)

Data Sheets:

Each training dataset includes a data sheet documenting:

- Data sources and collection methodology
- Composition (video sources, annotation schema, class distribution)
- Preprocessing steps
- Known biases or gaps
- Recommended use and limitations

Model Change Log:

Every model retrain and promotion includes:

- **Who**: Engineer or automated pipeline that initiated the change
- **What**: Specific changes (new training data, architecture change, hyperparameter tuning)
- **When**: Timestamp (UTC)
- **Why**: Trigger for change (accuracy drift, new annotated data, bug fix, improvement request)

- **Metrics**: Before/after accuracy comparison
- **Approval**: Production promotions include approver identity

Appendix A: Use Case Feasibility & Accuracy Matrix

#	Use Case	Feasibility	Accuracy Target	Detection Method	Priority
1	Man down	Yes	90-95%	Pose estimation + behavioral analysis	P1 (Pilot)
2	Tanks overfill/Vent	Conditional	88-93%	Anomaly detection + semantic segmentation	P2 (Rollout)
3	Machine Guarding	Yes	90-95%	Zone-based intrusion detection	P2 (Rollout)
4	Overcrowding	Yes	92-95%	Density estimation + zone counting	P1 (Pilot)
5	PPE — Goggles	Yes	90-95%	Multi-class object detection	P2 (Rollout)
6	PPE — Safety belt	Yes	90-95%	Person detection + torso ROI classification	P2 (Rollout)
7	PPE — Suit	Yes	90-95%	Full-body object detection	P2 (Rollout)
8	Over speeding	Conditional	90-95%	Vehicle detection + calibrated speed estimation	P2 (Rollout)
9	Using Fire hydrant hose	Conditional	88-93%	Object state detection	P2 (Rollout)
10	Dust cloud	Conditional	85-92%	Anomaly detection + temporal analysis	P2 (Rollout)
11	Fire & Smoke	Yes	95-97%	Dual-model (visual + temporal)	P1 (Pilot)
12	ANPR	Yes	95-98%	Two-stage: plate detection + OCR	P1 (Pilot)
13	Mobile phone usage	Yes	92-95%	Person detection + small object detection	P2 (Rollout)
14	Low Lux	Yes (camera req.)	85-92%	IR/thermal camera + augmented training	P1 (Camera spec)
15	Spillage	Yes	90-95%	Semantic segmentation + anomaly	P2 (Rollout)

				detection	
16	PPE — Shoes	Conditional	88-93%	Person detection + foot ROI classification	P2 (Rollout)

P1 (Pilot) : PPE (helmet/vest representative), Fire & Smoke, Man Down, ANPR, Overcrowding — highest impact, most mature models.

P2 (Rollout) : All remaining use cases deployed during rollout phase with iterative accuracy improvement.

Conditional : Feasible with dedicated cameras, site-specific training data, and iterative fine-tuning. Accuracy may require 2-3 improvement cycles to reach target.

Appendix B: Platform Comparison & Selection Rationale

Edge Inference Platform

Platform	Recommendation	Strengths	Limitations
NVIDIA Metropolis + DeepStream	**Selected**	Industry standard, 1000+ partner ecosystem, DeepStream handles multi-stream at scale, TAO for transfer learning, NIM for microservices	NVIDIA GPU lock-in
Intel OpenVINO	Alternative	CPU-based inference (no GPU needed), lower hardware cost	Lower throughput per unit, limited to Intel hardware
AWS Panorama	Alternative	Managed edge service, AWS ecosystem integration	AWS lock-in, higher recurring cost, less flexibility
Custom PyTorch/TensorFlow on edge	Alternative	Full control, no vendor dependency	Higher development effort, no managed pipeline tooling

Rationale : NVIDIA Metropolis is the de facto standard for industrial video analytics at scale. The DeepStream SDK handles multi-stream processing efficiently, TAO Toolkit enables rapid model fine-tuning, and the 1000+ partner ecosystem (including Wobot) ensures access to pre-trained industrial models. The GPU investment pays for itself in throughput per watt.

Cloud ML Platform

Platform	Recommendation	Strengths	Limitations
GCP Vertex AI	**Selected**	AutoML + custom training, model registry, monitoring, edge export, BigQuery integration, Gemini for conversational analytics	GCP ecosystem dependency
AWS SageMaker	Alternative	Mature ML platform, broad model support, Panorama edge integration	No Gemini equivalent for conversational analytics
Azure ML	Alternative	Enterprise integration, Azure ecosystem	Weaker edge deployment story for NVIDIA hardware
Self-hosted MLflow + Kubernetes	Alternative	Full control, no cloud lock-in	Higher operational overhead, no managed training infrastructure

Rationale : GCP Vertex AI provides the strongest integration path: Vertex AI for training → BigQuery for analytics → Gemini for conversational AI → Cloud Composer for orchestration. This is the InsightLens stack, proven in tvameva's analytics practice. The Mumbai region (asia-south1) satisfies data residency requirements.

Mobile App Framework

Framework	Recommendation	Strengths	Limitations
React Native	**Selected**	Cross-platform (iOS + Android), large ecosystem, shared codebase with React dashboard, strong push notification support	Performance slightly below native for complex animations
Flutter	Alternative	Google ecosystem alignment, excellent performance, single codebase	Smaller ecosystem for enterprise components, different language (Dart) from dashboard (React/JS)
Native (Swift + Kotlin)	Alternative	Best performance, full platform API access	Double development cost (two codebases), longer timeline

Rationale : React Native shares the JavaScript/TypeScript ecosystem with the React dashboard, enabling code reuse for shared components (alert models, authentication, API clients). Cross-platform delivery from a single codebase meets the 10-12 week timeline constraint.

tvameva.ai — AI-native. Outcome-anchored.

This document is confidential and proprietary to tvameva.ai. It may not be disclosed to third parties without prior written consent.